

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (CSE) (Credit System) Degree Examinations**

May - June 2023

20CSE81 – BIG DATA ANALYTICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I and Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Provide a comprehensive explanation of how the management of data has evolved over time, identifying and discussing the three waves of data management.	10	L*2	1	1
	b) Explain the cycle of Big Data Management using a diagram, and elaborate on the role of RDBMS in a Big Data environment.	10	L2	1	2
2.	a) Explain the examples of real time analytics. Determine the system capability requirements for real time analytics.	10	L2	1	1
	b) Explain design principles of any three layers of Big data stack using its block diagram.	10	L2	1	1
3.	a) Explain different examples for sources of both machine generated and human generated structured big data.	10	L2	1	1
	b) Provide a detailed comparison of the features of Traditional and Advanced Analytics, highlighting the differences between the two. Explain the role of CMS in a Big Data environment, including its functions and benefits.	10	L2	1	1
Unit – II					
4.	a) Provide a detailed description of the various features of the PostgreSQL relational database. You should also explain the extension capabilities of the PostgreSQL database, including its support for different data types.	10	L2	2	1
	b) Describe various NOSQL features of MongoDB database platform and strengths of CouchDB document database.	10	L2	2	1
5.	a) Provide a detailed description of the advantages and disadvantages of both SQL and NoSQL databases. Your answer should include the features that make each type of database unique and suitable for certain types of applications. Highlight the limitations and drawbacks of each type of database.	10	L2	2	1
	b) Identify various key step by step instructions of map reduce functions with the help of a data flow diagram.	10	L2	2	2
6.	a) Elaborate various attractive features of non-relational databases.	10	L2	4	1
	b) Explain various optimization techniques to improve the reliability and performance of MapReduce tasks.	10	L2	2	2
Unit – III					
7.	a) What are the various components of YARN and what are its two major services within the Hadoop ecosystem? Explain	10	L2	3	2
	b) What are the salient features of Hive and how does it utilize its three mechanisms for data organization? Explain in detail.	10	L2	3	1
8.	a) Explain various examples of basic analytics and advanced analytics in big data, and how do they differ from each other?	10	L2	5	2
	b) Describe fundamental architecture of HDFS with suitable block diagrams.	10	L2	4	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
